Elliott Hedman

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Project Portfolio

https://www.buildempathy.com/portfolio

Education

PH.D. | APRIL 2014 | MASSACHUSETTS INSTITUTE OF TECHNOLOGY

- Degree: Psychophysiology + AI + User Research
- Dissertation: "Thick Psychophysiology"
- Advisor: Rosalind Picard

B.S. | APRIL 2008 | UNIVERSITY OF COLORADO, BOULDER

- Major: Electrical Computer Engineering
- Minor: Leadership

Research Impact Summary:

Experienced UX researcher and AI-driven product strategist with a Ph.D. from MIT in Psychophysiology, AI, and User Research. After working at GoogleX and IDEO, I founded mPath to bring rigorous, advanced UX research methodologies to industry. My expertise in mixed-methods research, psychophysiology, and iterative prototyping has driven multi-million-dollar product decisions in digital education, interactive storytelling, and retail innovation.

Beyond research, I have an engineering background, building and deploying AI-driven applications including tools used by over 20,000 users. I specialize in bridging AI, behavioral science, and user-centered design to create impactful, research-driven experiences.

Founder & Lead UX Researcher, mPath ltd. (2013–Present):

As Founder & Principal Researcher at mPath, I built and led a UX research lab working with Fortune 500 companies and AI-driven products. Partnering directly with company leadership, I established UX research teams, integrated AI into research, and drove multi-million-dollar business impact. I collaborate with CEOs, VPs, and executive teams to establish UX research practices, define strategic UX goals, and drive data-informed innovation.

Key Achievements:

- Transformed UX research teams & scaled research practices at companies like The LEGO Group, McGraw Hill, Lowe's, and Curriculum Associates.
- Pioneered AI-driven UX methodologies, including AI-assisted eye tracking and psychophysiology
- Influenced multi-million-dollar product strategies, shaping global digital education, interactive play, and retail innovation.
- Mentored and scaled UX research teams, building research capacity across engineering, data science, and design teams.

Example Projects at mPath:

MCGRAW HILL – UX RESEARCH FOR THE FUTURE DIGITAL CLASSROOM (2022-2024)

- **Goal:** Redefine McGraw Hill's post-COVID digital learning strategy to drive adoption and engagement.
- **Approach:** Conducted iterative prototype testing with students in classrooms and focus groups with teachers to identify usability barriers and engagement drivers.
- **Key Insights:** New teachers prefer plug-and-play content; balanced reward systems improve student self-efficacy.
- **Impact:** Research shaped the company's digital roadmap, driving full adoption. Insights influenced high-impact releases for struggling readers and a 2025 platform launch (9M+ students).
- **Leadership:** Partnered with executive leadership, engineers, and instructional designers to define McGraw Hill's AI-driven UX research roadmap. Scaled research impact across departments, mentoring teams to move from usability testing to high-impact, data-driven UX strategy.

THE LEGO GROUP – TRANSFORMING LEGO BOOKS VALUE PROPOSITION (2021-2023)

- **Goal:** Reposition LEGO books from passive add-ons to interactive storytelling tools that enhance post-build play.
- **Approach:** Led a 6-month cross-functional UX research initiative with design, content, and data science teams, applying AI-assisted eye tracking and psychophysiology sensors across 42 prototype iterations.
- **Key Insights:** Children prioritize bricks over books; story-driven play thrives after building, making books the most valuable post-build.
- **Impact:** Research reshaped LEGO's publishing strategy, leading to Play Stories, one of LEGO's most successful products. Findings directly influenced global publishing and creative direction.
- **Leadership:** Led an R&D team of 7, mentoring designers and writers in UX research. Pioneered in-home, researcher-free methodologies to reduce observer bias. Insights were so influential that the Publishing Director pivoted his career to focus on this work.

Example Projects continued:

CURRICULUM ASSOCIATES – BUILDING A UX DEPARTMENT FOR DIGITAL EDUCATION (2016-2018)

- **Goal:** Transform i-Ready from a low-engagement digital tool into a student-centered, research-driven platform.

- **Approach:** Conducted naturalistic observations in homes and schools, applying psychophysiology research to measure engagement, followed by rapid iterative prototyping to refine learning experiences.

- **Key Insights:** Gamification (avatars & coins) failed to drive engagement—interactive, feedback-driven learning was far more effective.

- **Impact:** Redesign scaled i-Ready from 2M to 13M+ students, making it the industry leader. Teachers now cite i-Ready's UX approach as best practice: *"You should do it like i-Ready."*

- **Leadership:** Built and led Curriculum Associates' first cross-functional UX research program, collaborating with curriculum developers, data scientists, and engineers to transform i-Ready into a research-driven platform. Mentored and managed 3 user researchers.

LOWE'S HARDWARE – REINVENTING THE RETAIL EXPERIENCE (2013)

- **Goal:** Improve sales and customer decision-making for Wet-Dry Vacuums through human-centered retail design.

- **Approach:** Working with retail innovation, industrial design, and business teams, I conducted in-store usability testing and naturalistic shopper observations, followed by rapid iterative prototyping to optimize product displays and interactions.

- **Key Insights:** Customers were overwhelmed by choice; hands-on experience drove confidence more than text-based descriptions.

- **Impact:** Redesign led to a 7% increase in sales and secured a patent (US 10,080,475 B2) for interactive product displays. Findings influenced Lowe's shift toward UX-driven retail strategy.

- **Leadership:** Advised VP of Innovation on human-centered design adoption. Led a team of 5, mentoring researchers in prototyping & UX research methodologies.

Additional Research Projects

2024	Toddlers Can Read	Prototype and Design Parent App
2021	ClassCraft	Designing teacher onboarding for behavioral management [2]
2021	Kingsford	Understanding the emotional experience of buying charcoal
2021	Outschool	Understanding the effectiveness of digital lessons
2020	Education.com	Designing educational games for K-3 [3]
2018	Starbucks	Designing a better employee experience
2016	Colgate	Understanding the emotional difference brushing teeth between Hispanic and North American cultures
2015	EatWith	Improving user retention
2015	Taco Bell	Understanding the emotional experience of eating out
2015	Diagnostics for All	Helping cancer patients identify their diagnosis
2015	Sphero	Engaging children with creative play and robotics
2014	Estée Lauder	Distinguishing smash box products in retail
2014	Best Buy	Reducing shopper stress
2014	Crown	Improving interactive person-to-person sales
2014	Kaiser Permanente	Designing a better experience for diabetes patients
2013	The LEGO Group	Designing digital manuals to help children with difficult builds [9]
2012	New World Symphony	Making concerts engaging for first-comers [9]
2011	Hasbro	Designing game instructions to be more accessible to multi-language families [12]
2011	Google	Reducing stress in self-driving cars
2011	Blue Man Group	Improving the audience experience

Previous Work

2008 – 2014	MIT Media Lab	Research Assistant
2012-2013	IDEO	Design Researcher
2011	Google	UX Researcher

Skills

Qualitative Research:

Contextual Inquiry, Diary Studies, Moderated & Unmoderated Usability Testing, Ethnographic Research, Behavioral Interviews

Quantitative Research:

A/B Testing, Large-Scale Survey Research, Eye Tracking, Psychophysiology (EDA, Heart Rate, Stress Biometrics), **Inferential Statistics, Bayesian Modeling, Experimental Design**

Programming & Research Tooling:

Prototype Development: Unity, C#, JavaScript, App Development
Research Tooling: Google Cloud, Python/Matlab (for Data Analysis), SQL
Machine Learning & AI for UX: Behavioral Modeling, LLM Prompt Engineering, AI-Driven UX Research

Data & Behavioral Analytics:

Eye Tracking & Gaze Analysis, Attention Mapping, Product Usage Metrics, Interaction Logging, Predictive Behavioral Modeling, Machine Learning

Prototyping & Testing:

Iterative Prototyping, Co-Creation Workshops, Rapid Concept Validation

Leadership & Strategy:

Research Team Mentorship, UX Research Evangelism, Executive & Stakeholder Influence, UX Roadmap Development

Publications

[1] Hedman, E., & Ana Gilbertson. (2023, June 28). *5 Ways Digital Books Can Empower Play: A Psychophysiological Design Study*. ISTELive.

[2] Hedman, E. (2023a, March 11). 3 Pillars of Teacher Adoption for Edtech. *Buildempathy.com*.

[3] Hedman, E. (2023b, May 15). *A Rubric for Making Engaging Educational Games*. Wonder.Io.

[4] Hedman, E. (2023c, December 18). Why Kids Work Hard at Fortnite, Not Frog and Toad. *Medium*.

[5] Hedman, E. (2021). Motivating Struggling Readers to Mentally "Show Up" with Wonder Stories. *ELO 2021 Conference and Festival: Platform (Post?) Pandemic*.

[6] Hedman, E., Schoen, S. A., Miller, L. J., & Picard, R. (2020). Wireless Measurement of Sympathetic Arousal During in vivo Occupational Therapy Sessions. *Frontiers in Integrative Neuroscience*, *14*.

[7] Hedman, E. (2018, March). 7 Ways to Help Kids Feel they're Learning. *Medium*, 1.

[8] Hedman, E., Zanchi, C., & Hedman, G. (2018). Effective Feedback Depends on the Emotional Weight of a Problem. *AERA 2018 Annual Meeting*.

[9] Hedman, E. (2014). Thick Psychophysiology for Empathic Design. *Massachusetts Institute of Technology*, *2008*. https://dspace.mit.edu/handle/1721.1/91436

[10] Hedman, E., Miller, L. , Schoen, S., Nielsen, D., Goodwin, M., & Picard, R. (2012). - Measuring autonomic arousal during therapy. In J. Brassett, J. McDonnell, & M. Malpass (Eds.), *Proceedings of 8th International Design and Emotion Conference : Vol. London*.

[11] Hedman, E. , Quigley, K. S., & Picard, R. W. (2012). USING RANKED SCRS IN AMBULATORY MEASUREMENTS: A NEW APPROACH TO DISTINGUISHING REAL-WORLD SALIENT EVENTS. *PSYCHOPHYSIOLOGY*, *49*, S118–S118.

[12] Hedman, E. (2011). The Frustration of Learning Monopoly: The Emotional Tension of Entering a New Game Encounter. *The Ethnographic Praxis in Industry Conference, September 18-21*, *Boulder*.

[13] Hedman, E. (2010). *In-situ Measurement of Electrodermal Activity During Occupational Therapy: Vol. Masters*.

[14] Fletcher, R., Dobson, K., Goodwin, M., Eydgahi, H., Wilder-Smith, O., Fernholz, D., Kuboyama, Y., Hedman, E., Poh, M., & Picard, R. (2010). iCalm: Wearable sensor and network architecture for wirelessly communicating and logging autonomic activity. *Information Technology in Biomedicine, IEEE Transactions On*, *14*(2), 215–223.

Presentations

Hedman, E. (2024). Level up Your User Research: Why EdTech Needs to Co-Create. *SXSW EDU*.

Hedman, E. (2020c). Transforming reading into a multisensory experience. *Star Institute*.

Hedman, E. (2020b). Using emotional sensors to understand your users. *GitHub Universe*.

Hedman, E. (2020a). Books 2.0: Why Reading Needs to Go Digital. *SXSW EDU*.

Hedman, E. (2019). The Emotional Internet of Things. *Coffee with Mr. IoT. Deloitte*.

Hedman, E. (2019). The Death of Curiosity. *SXSW*.

Hedman, E. (2018b). Why "Just Add Games" Doesn't Boost Engagement. *SXSW EDU*.

Hedman, E. (2018a). Emotional Sensors to Uncover User Needs. *The Conference*.

Hedman, E. (2015). The Importance of Parents in Play. *SXSW EDU*.

Hedman, E. (2012). User Research with Psychophysiological Sensors. Design Research Conference. *IIT Institute of Design*.

Hedman, E. (2011). Improving the Emotional Experience of Kids with Autism. *TedX*.

Press

Learning Designers Call for More User Testing of Edtech Products and Teaching Materials. Edsurge.com. 2024. https://www.edsurge.com/news/2024-03-19-learning-designers-call-for-more-user-testing-of-edtech-products-and-teaching-materials

Motivating Children to "Show up to Reading". Scholastic.edu. 2021. https://edublog.scholastic.com/post/motivating-children-show-reading

Can we make ourselves more empathetic? 100 years of research still has psychologists stumped. Popular Science. 2022. https://www.popsci.com/story/science/popsci-archive-empathy-machine/

Wonder Stories: Proven to engage child readers like digital games. Patch.com. 2020. https://patch.com/illinois/downersgrove/wonder-stories-proven-engage-child-readers-digital-games

Wearable device reveals consumer emotions. MIT News. 2017. https://news.mit.edu/2017/wearable-device-reveals-consumer-emotions-0712

Beyond Solution: Science of Emotion. Wired Japan. 2015. https://news.mit.edu/2017/wearable-device-reveals-consumer-emotions-0712

How Sensors That Test Our Stress Could Revolutionize Product Design. WIRED. 2014. https://www.wired.com/2014/12/sensors-test-stress-revolutionize-product-design/

Can Data Teach Us Empathy? Information Week. 2015. https://www.informationweek.com/it-leadership/can-data-teach-us-empathy-#close-modal

How Wearable Tech Could Improve Your Mental Health. Live Science. 2014. https://www.livescience.com/44516-future-of-smart-wristbands.html

Select 10 Up-and Coming Designers. Metropolis. 2014.

The Quantified Spouse Movement Has Couples Tracking Weight, Sleep And Even Orgasms To Find Bliss. Huffington Post. 2013. https://www.huffpost.com/entry/quantified-spouse-movement_n_2567459

Patents

Hedman, E. , Chi, L. T., Wheeler, A. J. Display Method. (U. S. Patent No. 8957916B1). U.S. Patent and Trademark Office. Displaying messages on Google Glass that are more easily readable by using augmented reality to determine best place to display a message.

Grey, J., Cooley, A., Ferris, B. Murray, C., Hedman, E., Slate, B. Indicators of cleaning effectiveness of vacuum cleaners and systems to facilitate comparisons of cleaning effectiveness of vacuum cleaners (U. S. Patent No. US10080475B2). U.S. Patent and Trademark Office.